Food Trends Understanding Customer Preferences in FB

Submitted by:

Tayyaba Haroon

DECLARATION

I, Tayyaba Haroon, hereby declare that the project report titled "**Food Trends Understanding Customer Preferences in FB**" is my original work carried out by me as part of my Infosys Springboard Virtual Internship. This report has not been submitted to any other institution or organization for the award of any degree, diploma, or certification.

All sources of information used in this project have been properly acknowledged. The work presented in this report is based on the data analysis performed by me using Microsoft Power BI, and any assistance received during the preparation of this report has been duly recognized.

I further declare that the results presented in this report are true to the best of my knowledge and belief.

ABSTRACT

This project presents a comprehensive analysis of restaurant performance and customer behavior using Microsoft Power BI. The study utilizes a structured dataset containing information such as cuisine type, city, average ratings, customer votes, and engagement indicators. The primary objective is to transform raw data into meaningful insights that support data-driven decision-making in the food service industry.

The analysis begins with data cleaning and transformation using Power Query Editor to ensure accuracy and consistency. Sentiment analysis is conducted using rating-based inference to evaluate overall customer satisfaction. Product insights are derived to identify top-performing and underperforming cuisine categories based on ratings and engagement levels. Customer segmentation is performed using behavioral indicators, classifying users into Loyal Fans, Critics, and Silent Users to understand interaction patterns.

Additionally, a Strategy and Innovation framework is developed to identify “Hidden Gem” restaurants—highly rated establishments with low visibility. Interactive dashboards are created to visualize trends and relationships across multiple dimensions. The findings demonstrate that combining satisfaction metrics with engagement indicators provides

TABLE OF CONTENTS

1. Project Objectives
2. Methodology
    2.1 Data Collection
    2.2 Data Preparation and Cleaning
    2.3 Data Transformation
    2.4 Sentiment Analysis
    2.5 Product and Performance Analysis
    2.6 Customer Segmentation
    2.7 Analysis of Customer Preferences
    2.8 Regional Analysis
    2.9 Strategy and Opportunity Identification
3. Dashboard Development
    3.1 Sentiment Analysis Dashboard
    3.2 Product Insights Dashboard
    3.3 Customer Segmentation Dashboard
    3.4 Strategy & Innovation Dashboard
4. General Observations
5. Business Implications
6. Conclusion

1.PROJECT OBJECTIVES

The primary objective of this project is to analyze food industry data to extract meaningful insights related to customer preferences, restaurant performance, engagement patterns, and market trends using modern data analytics techniques. The study focuses on transforming raw restaurant data into actionable insights through interactive dashboards developed in Power BI.

This project aims to demonstrate how data visualization and analytical modeling can support informed decision-making within the food and beverage sector. By examining rating behavior, customer engagement levels, sentiment patterns, and product performance, the analysis seeks to identify both high-performing segments and areas requiring strategic improvement.

The specific objectives of the project are as follows:

- To analyze restaurant data to understand customer behavior and engagement patterns.
- To identify trends in cuisine popularity, ratings, and performance indicators.
- To perform sentiment analysis to evaluate customer satisfaction levels.
- To segment users based on behavioral indicators derived from engagement and ratings.
- To detect underperforming categories and hidden opportunities within the dataset.
- To develop interactive dashboards that present complex insights in a clear and accessible format.
- To support data-driven decision-making through visual analytics.
- To demonstrate the practical application of business intelligence tools in real-world scenarios.

Through these objectives, the project illustrates how analytics can help businesses better understand consumer behavior and optimize strategic planning in a competitive food market.

2. METHODOLOGY

The methodology adopted for this project follows a structured data analytics pipeline that converts raw restaurant data into meaningful insights. The process includes data collection, data preparation, analysis, modeling, dashboard development, and interpretation.

2.1. Data Collection

The dataset used in this study consists of restaurant-level information relevant to the food and beverage industry. The data includes variables such as restaurant name, city, cuisine category, average rating, number of votes, pricing indicators, and other performance metrics.

Since direct customer records were not available, customer behavior was inferred from interaction data such as ratings and engagement levels. This approach allows behavioral analysis using proxy indicators.

	Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column
1	Restaurant Name	Dining Rating	Delivery Rating	Dining Votes	Delivery Votes	Cuisine	Place Name	City
2	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
3	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
4	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
5	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
6	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
7	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
8	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
9	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
10	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
11	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
12	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
13	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
14	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
15	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
16	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
17	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
18	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
19	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
20	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
21	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
22	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
23	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad
24	Doner King	3.9	4.2	39	0	Fast Food	Malakpet	Hyderabad

2.2. Data Preparation and Cleaning

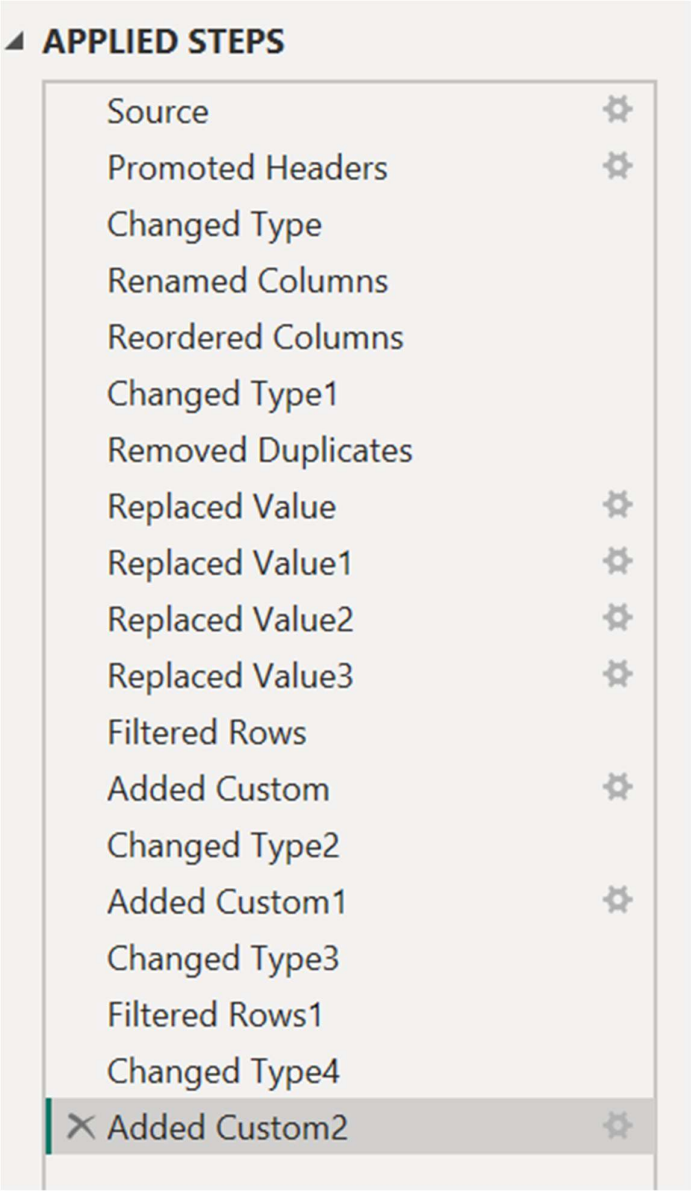
Before analysis, the dataset underwent extensive preprocessing using Power Query Editor in Power BI.

The data preparation process included:

- Removing duplicate records
- Handling missing or null values
- Standardizing text formats
- Converting columns to appropriate data types
- Eliminating inconsistencies
- Filtering irrelevant entries

Null values were carefully handled to prevent distortion of analytical results. Where appropriate, missing values were removed or replaced using logical assumptions based on context.

This step ensured that the dataset was clean, reliable, and suitable for analysis.

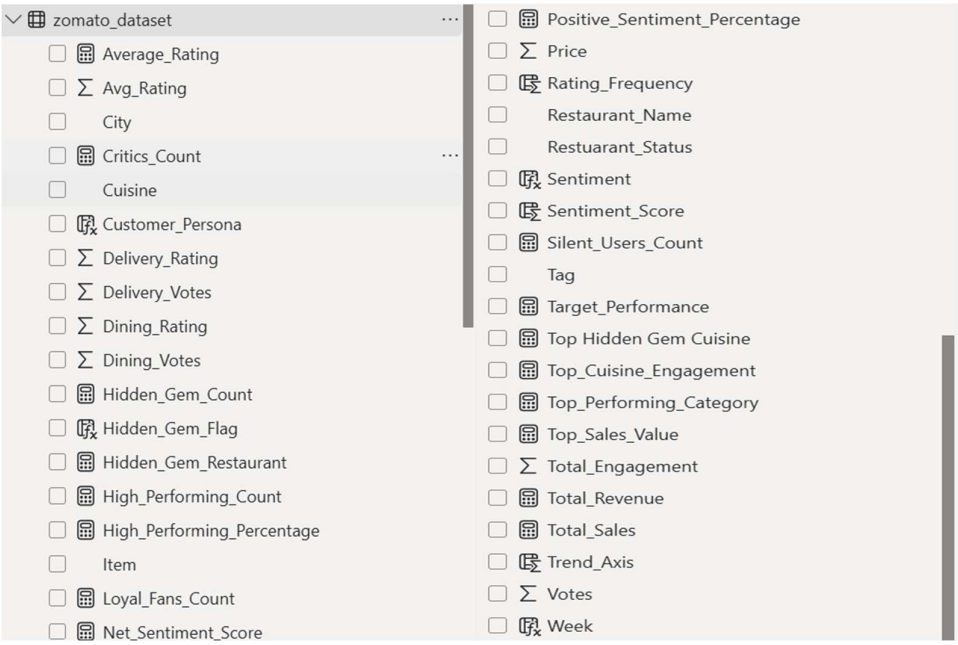


2.3. Data Transformation

Additional calculated columns and measures were created to enhance analytical capability. These transformations included:

- Engagement indicators derived from votes
- Rating frequency metrics
- Sentiment classification variables
- Performance indicators
- Aggregated measures for dashboards

Power BI's DAX functions were used extensively to create these derived fields. Transforming the raw data into analytical variables enabled deeper insights into restaurant performance and customer behavior.



2.4. Sentiment Analysis

Sentiment analysis was conducted using rating-based inference due to the absence of textual reviews.

Ratings were used as a proxy to classify sentiment into:

- Positive
- Neutral
- Negative

A Net Sentiment Score was computed to represent overall customer perception.

This analysis helped identify:

- Customer satisfaction trends
- Areas of strong performance
- Segments with negative perception
- Opportunities for service improvement

Sentiment analysis plays a critical role in understanding consumer attitudes toward restaurants and cuisine categories.

2.5. Product and Performance Analysis

Product insights were derived by examining cuisine categories, sales indicators, engagement levels, and ratings.

Key analytical questions addressed include:

- Which cuisines perform best in terms of popularity and satisfaction?
- Which categories attract the highest engagement?
- Which products show declining performance?

Visualizations such as bar charts and KPI cards were used to highlight top-performing categories and underperforming segments.

This analysis enables businesses to prioritize offerings that maximize customer satisfaction and revenue potential.

2.6. Customer Segmentation

Since individual customer data was unavailable, segmentation was based on behavioral indicators inferred from interaction patterns.

Three primary variables were used:

- Average Rating (satisfaction)
- Votes (engagement)
- Sentiment score

Based on these indicators, users were categorized into three personas:

1. Loyal Fans

Users who consistently provide high ratings and demonstrate strong engagement.

2. Critics

Users who give relatively lower ratings or express dissatisfaction.

3. Silent Users

Users who interact with restaurants but provide limited feedback or engagement.

This segmentation provides valuable insight into how different user groups influence restaurant performance.

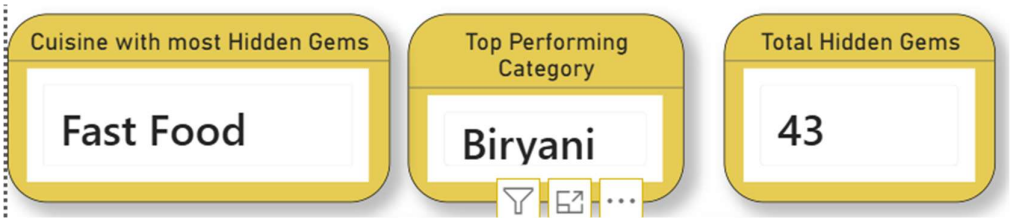
```
Customer_Persona =
SWITCH(
TRUE(),
zomato_dataset[Avg_Rating] >= 4 && zomato_dataset[Dining_Votes] > 100, "Loyal Fans",
zomato_dataset[Avg_Rating] < 3.5, "Critics",
"Silent Users"
)
```

2.7. Analysis of Customer Preferences

The relationship between customer personas and cuisine categories was examined to determine product preference patterns.

Findings indicated that some cuisines attract strong loyalty while others receive passive interaction. For example, popular fast-food outlets often experience high engagement but may also have a large proportion of silent users.

Understanding these patterns helps businesses tailor marketing strategies and improve customer experience.

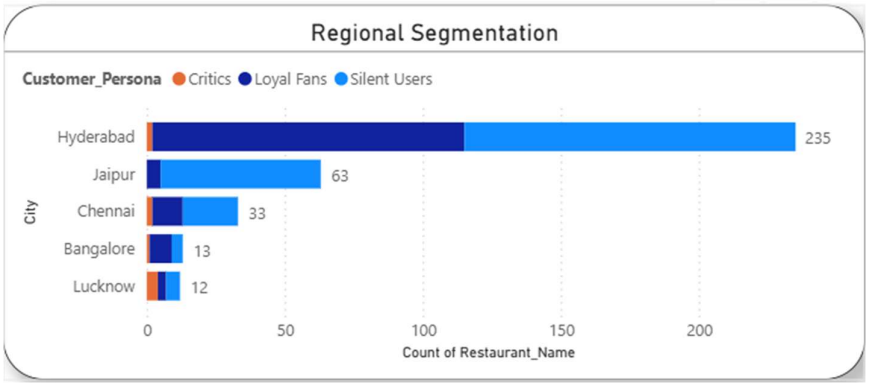


2.8. Regional Analysis

Geographic differences in customer behavior were analyzed across cities.

Regional segmentation revealed variations in engagement levels, satisfaction, and persona distribution. Certain cities demonstrated strong loyalty, while others showed a higher concentration of passive users.

These insights can guide location-specific strategies such as targeted promotions or service improvements.



2.9. Strategy and Opportunity Identification

One of the most significant analytical outcomes of the project was the identification of "Hidden Gem" restaurants.

Hidden Gems are defined as restaurants with:

High customer satisfaction (high ratings)

Low visibility or engagement (low votes)

These establishments represent untapped potential that could achieve higher success with improved marketing or exposure.

A Restaurant Opportunity Matrix was developed to classify restaurants into strategic quadrants based on engagement and satisfaction.

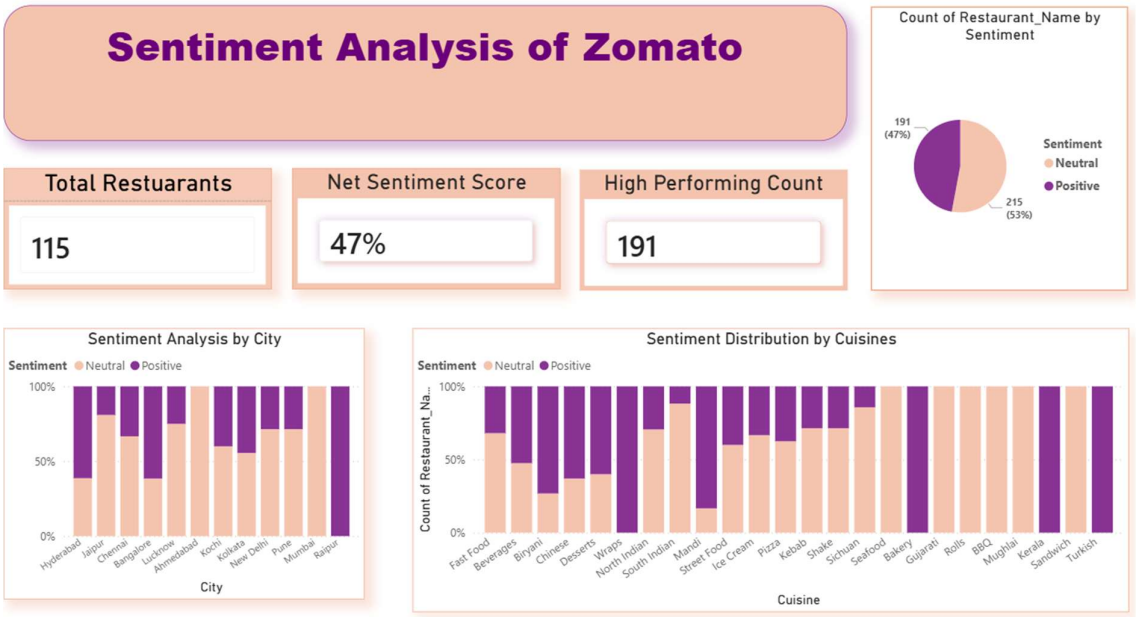
```
Hidden_Gem_Flag =
IF(
    zomato_dataset[Avg_Rating] >= 4 &&
    zomato_dataset[Votes] < 50,
    "Hidden Gem",
    "Other"
)
```

3. DASHBOARD DEVELOPMENT

Interactive dashboards were created in Power BI to visually communicate insights.

The dashboards include:

3.1.Sentiment Analysis Dashboard



The sentiment analysis dashboard evaluates overall customer perception of restaurants using rating-based sentiment indicators. Since textual reviews were not available, sentiment was inferred from numerical ratings and engagement patterns.

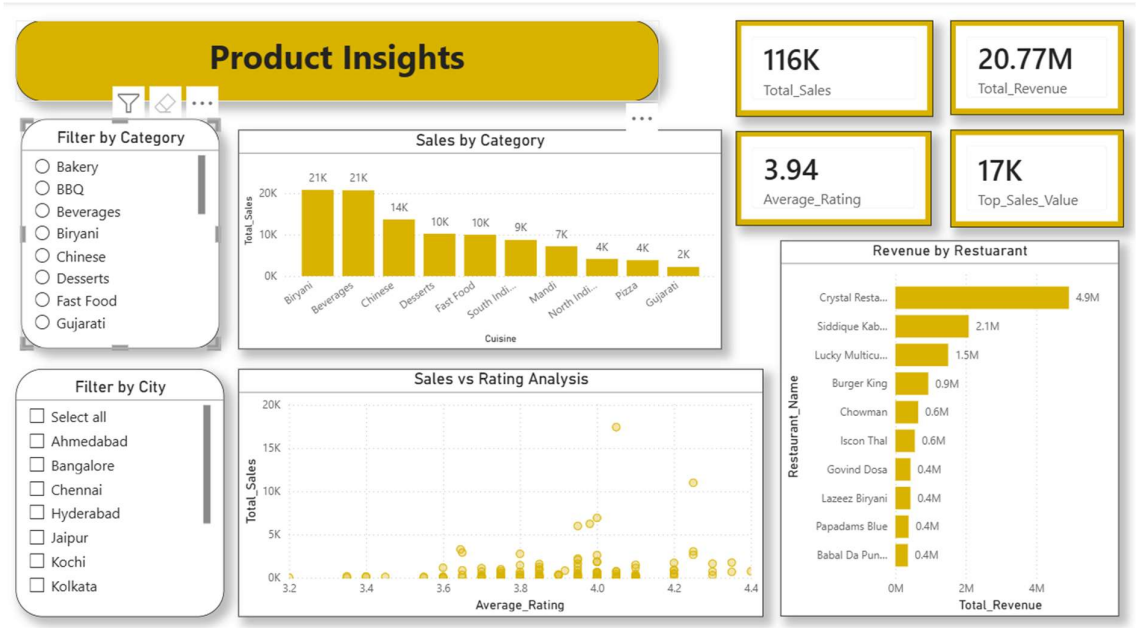
Key Findings

- The majority of restaurants exhibit positive sentiment, indicating generally high customer satisfaction across the dataset.
- A proportion of restaurants fall into neutral sentiment categories, suggesting areas requiring service or quality improvements.
- High sentiment scores are strongly associated with higher average ratings, confirming the reliability of ratings as a proxy for customer satisfaction.
- Certain cuisine categories consistently demonstrate stronger positive sentiment, reflecting higher customer approval.
- Variations in sentiment across cities indicate that customer experience differs geographically.
- Restaurants with high engagement (votes) tend to maintain more stable sentiment patterns due to larger feedback samples.

- Lower engagement restaurants often show volatile sentiment, suggesting limited feedback may distort perception.

Overall, the sentiment analysis indicates that while the food industry segment under study maintains generally positive customer perception, targeted improvements are required for lower-performing segments.

3.2.Product Insights Dashboard



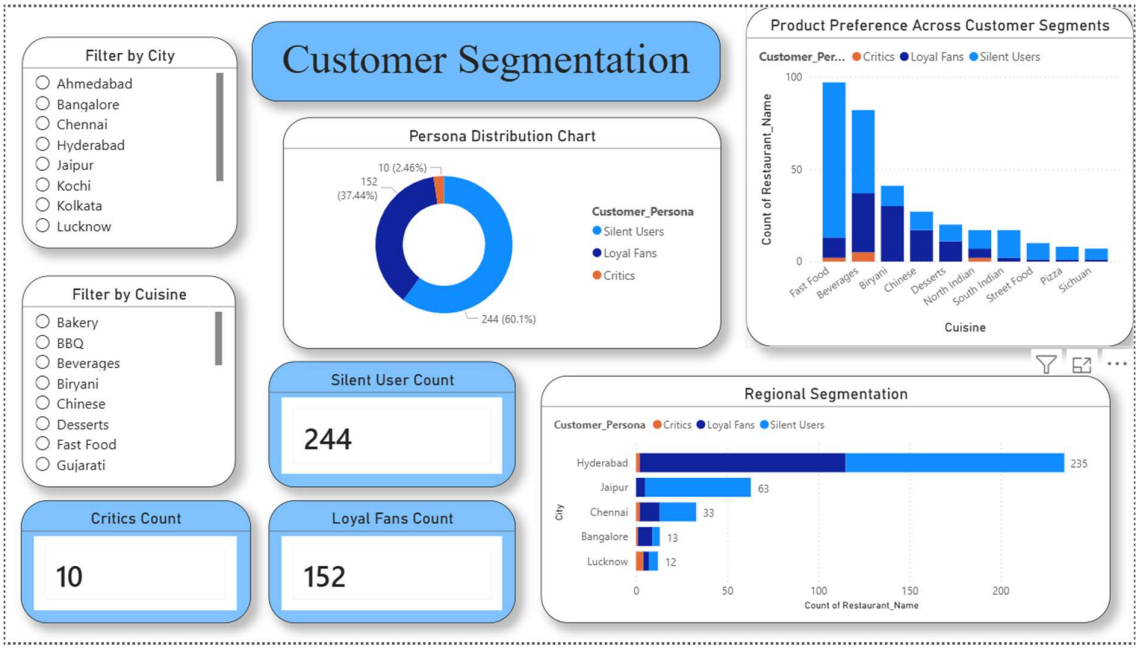
The product insights dashboard focuses on identifying performance trends across different cuisine categories using ratings, engagement levels, and popularity indicators.

Key Findings

- Certain cuisines such as Biryani, Beverages and Chinese emerge as top-performing categories, demonstrating strong customer approval and demand.
- Fast Food, despite high popularity, shows mixed performance due to variability in ratings and customer satisfaction.
- Some categories attract significant engagement but relatively lower ratings, indicating popularity without consistent quality perception.
- Emerging categories with moderate engagement but strong ratings represent growth opportunities.
- Underperforming cuisines display consistently lower average ratings, highlighting areas requiring product improvement or repositioning.
- Seasonal or trend-driven categories exhibit fluctuations in performance indicators.
- Customer preferences appear concentrated around a limited number of dominant cuisines.

These findings enable stakeholders to prioritize high-performing categories while implementing improvement strategies for weaker segments.

3.3.Customer Segmentation Dashboard



The customer segmentation analysis reveals distinct behavioral patterns in how users interact with restaurants.

Key Findings:

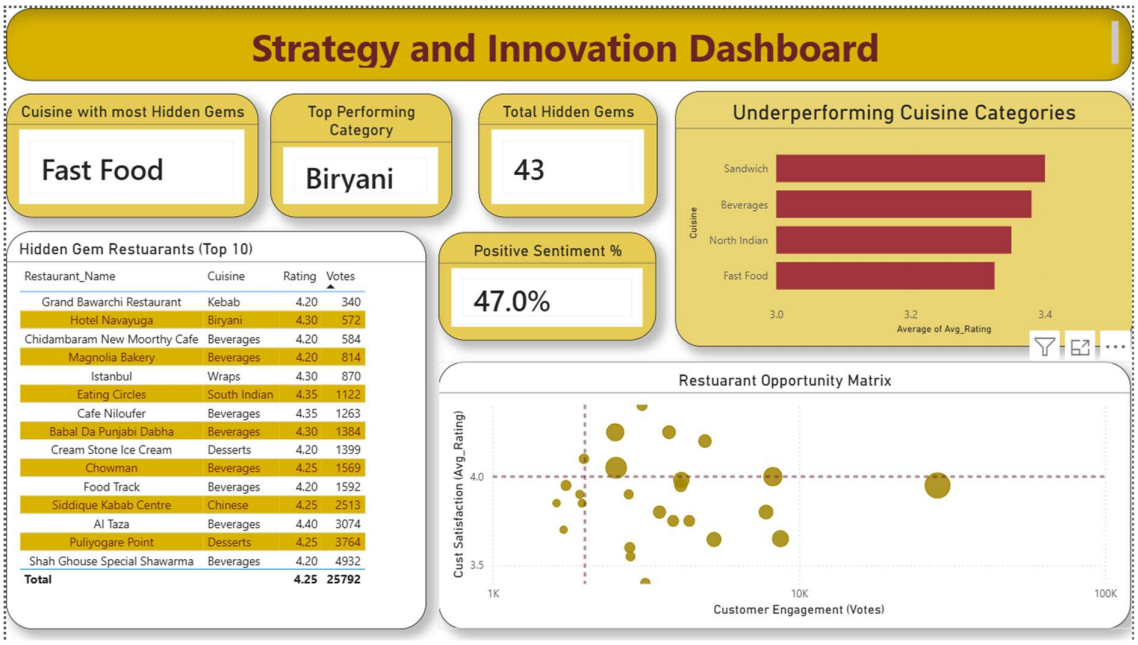
- Silent Users constitute the largest segment, indicating that many users engage with restaurants passively without actively providing feedback.
- Loyal Fans represent a significant group that consistently provides high ratings and positive engagement, contributing strongly to restaurant reputation.
- Critics form a relatively small but important segment that highlights dissatisfaction and potential service issues.
- Product preferences vary across segments, certain cuisines attract stronger loyalty while others see mostly passive engagement.
- Fast Food shows high interaction but comparatively lower loyalty, suggesting frequent consumption with limited satisfaction.
- Cuisines such as Chinese and Biryani demonstrate stronger Loyal Fan presence, indicating higher customer satisfaction.
- Regional differences are evident, with some cities showing higher positive engagement while others are dominated by Silent Users.

- Cities with large Silent User populations present opportunities for targeted engagement strategies.

Business Implications:

- Restaurants should focus on converting Silent Users into active supporters through loyalty programs and improved customer experience.
- Retention strategies should prioritize Loyal Fans to maintain positive brand perception.
- Feedback from Critics should be analyzed to address service gaps.
- Marketing and operational strategies should be tailored based on regional engagement patterns.

3.4.Strategy & Innovation Dashboard



Restaurant_Name	Cuisine	Rating	Votes
Grand Bawarchi Restaurant	Kebab	4.20	340
Hotel Navayuga	Biryani	4.30	572
Chidambaram New Moorthy Cafe	Beverages	4.20	584
Magnolia Bakery	Beverages	4.20	814
Istanbul	Wraps	4.30	870
Eating Circles	South Indian	4.35	1122
Cafe Niloufer	Beverages	4.35	1263
Babal Da Punjabi Dabha	Beverages	4.30	1384
Cream Stone Ice Cream	Desserts	4.20	1399
Chowman	Beverages	4.25	1569
Food Track	Beverages	4.20	1592
Siddique Kabab Centre	Chinese	4.25	2513
Al Taza	Beverages	4.40	3074
Puliyogare Point	Desserts	4.25	3764
Shah Ghouse Special Shawarma	Beverages	4.20	4932
Total		**4.25**	**25792**

The Strategy and Innovation Dashboard highlights performance gaps, growth opportunities, and areas requiring strategic attention within the restaurant ecosystem.

Key Findings:

- Several restaurants exhibit high ratings but low engagement, identifying them as Hidden Gems with strong potential for growth.
- The cuisine category with the highest concentration of hidden gems indicates uneven visibility within popular segments.
- Fast Food emerges both as a category with many hidden gems and as an underperforming category, suggesting mixed quality across establishments.
- The top-performing category demonstrates consistently high customer satisfaction and engagement, serving as a benchmark for success.

- Certain cuisine categories show lower average ratings, indicating areas where service quality or product offerings may need improvement.
- Hidden Gem restaurants represent untapped opportunities that could benefit significantly from increased promotion and visibility.
- The opportunity matrix shows that engagement and satisfaction are not always aligned, popularity does not guarantee quality.

Business Implications:

- Marketing efforts should prioritize promoting hidden gems to unlock their growth potential.
- High-performing categories should be leveraged for expansion and brand positioning.
- Underperforming segments require quality improvements and strategic intervention.
- Data-driven targeting can optimize resource allocation across categories and regions.
- Identifying and nurturing high-quality but low-visibility restaurants can improve overall market competitiveness.

Each dashboard integrates multiple visual elements such as:

- KPI cards
- Bar charts
- Donut charts
- Scatter plots
- Filters and slicers

Interactive features allow users to explore insights dynamically across categories, cities, and customer segments.

4. GENERAL OBSERVATIONS

The overall analysis of restaurant data reveals several important behavioral and performance patterns within the food industry.

Major Observations

- Customer satisfaction levels are generally high across the dataset, as reflected by predominantly strong average ratings.
- Engagement levels vary widely, indicating that visibility and popularity differ significantly among restaurants.
- A large number of restaurants receive relatively few votes despite strong ratings, suggesting limited awareness rather than poor performance.
- Silent users constitute a major portion of interactions, indicating passive consumption behavior where customers engage without providing extensive feedback.
- Loyal fans, while fewer in number, contribute disproportionately to positive sentiment and strong ratings.
- Critics represent a small but influential segment whose feedback highlights areas requiring improvement.
- Cuisine popularity does not always correlate with satisfaction — highly popular categories may still receive mixed reviews.
- Regional differences suggest that local market conditions influence customer behavior and preferences.
- Hidden gem restaurants demonstrate that high-quality offerings may remain under-recognized due to low exposure.
- The distribution of engagement across restaurants is highly uneven, with a few establishments receiving a large share of attention.

These observations highlight the importance of analyzing both satisfaction and engagement metrics simultaneously to obtain a comprehensive understanding of performance.

5. BUSINESS IMPLICATIONS

The insights derived from this analysis provide valuable guidance for strategic decision-making within the food and beverage industry.

Strategic Implications

1. **Marketing and Visibility Enhancement**

Restaurants identified as hidden gems represent high-quality offerings with low market visibility. Targeted promotional campaigns can significantly improve their performance.

2. **Customer Engagement Strategies**

The dominance of silent users suggests a need for initiatives that encourage feedback and interaction, such as loyalty programs, review incentives, or personalized communication.

3. **Product Optimization**

Underperforming cuisines should be evaluated for quality improvements, pricing adjustments, or repositioning to better align with customer expectations.

4. **Resource Allocation**

Businesses can allocate resources more efficiently by focusing on high-potential segments identified through data analysis.

5. **Geographic Strategy**

Regional variations in customer behavior indicate the need for location-specific marketing and service strategies.

6. **Customer Retention**

Loyal fans represent valuable long-term customers. Maintaining high service standards for this group is essential to sustain positive brand perception.

7. **Reputation Management**

Monitoring sentiment trends allows organizations to proactively address negative feedback and prevent deterioration of customer trust.

8. **Competitive Advantage**

Data-driven insights enable businesses to anticipate market trends and adapt more quickly than competitors relying solely on intuition.

6. CONCLUSION

This project demonstrates the critical role of data analytics and visualization in understanding customer behavior within the food and beverage industry.

By analyzing restaurant data through Power BI dashboards, valuable insights were obtained regarding cuisine popularity, engagement levels, satisfaction patterns, and strategic opportunities. The study highlights that customer decisions are influenced not only by product quality but also by visibility, engagement, and regional factors.

The identification of hidden gem restaurants underscores the importance of data-driven discovery in uncovering untapped potential. Additionally, the segmentation of users into behavioral personas provides a deeper understanding of interaction patterns.

Ultimately, the project emphasizes that data-driven decision-making is essential for competitiveness in modern markets. Businesses that leverage analytics to understand customer preferences can refine their offerings, strengthen relationships, and achieve sustainable growth.